Notes: Callen, Isaqzadeh, Long & Sprenger (AER, 2014)
Violence and Risk Preference: Experimental Evidence from Afghanistan

Contents

1 Big picture

- **Question.** Does exposure to political violence change economic risk preferences, and does recalling fearful events trigger different risk choices among people exposed to violence?
- **Core setting.** The paper studies Afghan civilians in a conflict environment, combining field experiments with administrative records of violent events from the International Security Assistance Force's Combined Information Data Network Exchange.
- **Main experimental tool.** Subjects complete two uncertainty-equivalent tasks that separately elicit utility under uncertainty and utility under certainty. The difference is called the **Certainty Premium**.
- **Main psychological tool.** Respondents are randomly assigned to a fear, happy, or neutral recollection prime. The fear prime asks respondents to recall a fearful episode, borrowing methods from psychology.
- **Main violence measure.** Violence exposure is measured administratively using geocoded insurgent attacks within a one-kilometer radius of interview location during April 2002–February 2010.
- **Main behavioral fact.** Subjects display a positive Certainty Premium: they value a sure payment more than its equivalent uncertain value. This violates expected utility and is consistent with certainty-specific preferences.
- **Main treatment result.** Fearful recollection increases the Certainty Premium specifically for individuals exposed to violence. Exposure alone does not mechanically shift preferences in the same way.
- **Interpretation of trauma.** The paper argues that violence may not permanently shift a person's baseline risk preference; rather, it changes the person's susceptibility to fear-triggered decision states.
- **Policy relevance.** If fear recall triggers higher demand for certainty, trauma-affected individuals may respond differently to insurance, savings, portfolio choice, conflict interventions, or policy framing.
- **Economic takeaway.** Risk preference is not a single stable object in conflict settings. What matters is the interaction between objective exposure, emotional recall, and the form of the risky choice.

2 Incremental contribution

2.1 What this paper adds relative to prior work

- **From correlation to combined experimental design.** Prior work often compared exposed and unexposed people after disasters or wars. This paper combines objective exposure data with randomized psychological primes, allowing the authors to separate background exposure from experimentally triggered fear recall.
- **From generic risk aversion to certainty-specific preferences.** Many studies estimate a single risk-aversion parameter. This paper introduces a field measure that distinguishes utility elicited under uncertainty from utility elicited at certainty, generating the Certainty Premium.
- **From self-reported exposure to geocoded administrative violence.** The paper uses precisely geocoded military records rather than relying only on retrospective self-reports, which can suffer from recall error, underreporting, or trauma-induced avoidance.
- **From laboratory psychology to conflict field economics.** The paper imports fear-recollection primes from psychology into a high-stakes field setting where trauma is widespread and economically relevant.
- **From permanent-preference interpretation to triggerability.** Earlier evidence could be interpreted as trauma permanently changing risk preferences. This paper shows that exposed individuals are particularly responsive to fear recall, suggesting a state-dependent triggerability mechanism.

- **From expected utility tests to non-EU mechanisms.** The two-task design tests the independence axiom and separates predictions of expected utility, cumulative prospect theory, disappointment aversion, and u–v preferences.
- **From broad trauma effects to actionable decision contexts.** The result is not just that violence matters; it specifies a behavioral margin–the demand for certainty–that may be relevant for insurance, savings products, policy communication, and post-conflict interventions.

Economic message: The incremental contribution is methodological and substantive. Methodologically, the paper creates a field-compatible way to isolate certainty preference. Substantively, it shows that violence exposure can make economic behavior more sensitive to emotional triggers, which is different from simply saying that violence makes people more or less risk averse.

3 Data, experimental design, and measurement

3.1 Survey sample and field context

- The study is conducted in Afghanistan after years of insurgent violence.
- The survey includes risk-preference modules embedded in a broader household survey of experiences, attitudes, beliefs, and demographics.
- The paper reports that 2,027 respondents were approached in 12 provincial centers, with 1,127 respondents consenting to the risk-preference protocol.
- The clean experimental sample used in the main tables includes 816 individuals with monotonic utility and no multiple switching in the risk tasks.
- Polling centers are used as primary sampling units and as geographic units for randomization and violence assignment.

Interpretation: The sample is not a standard student lab sample. The design is intended to measure economic preferences in an environment where exposure to violence is common and plausibly salient for decision making.

3.2 Multiple price list tasks

- Each respondent completes two tasks, each containing ten binary choices.
- In both tasks, Option B is a risky gamble with a probability q of winning 450 Afghanis and probability $1 - q$ of winning 0 Afghanis.
- In Task 1, Option A is an uncertain gamble: 50 percent chance of 450 Afghanis and 50 percent chance of 150 Afghanis.
- In Task 2, Option A is a certain payment of 150 Afghanis.
- The probability q in Option B increases from 0.1 to 1 across rows. The row where a respondent switches from Option A to Option B identifies an interval for the utility value of 150 Afghanis.

Interpretation: The two tasks are identical in their Option B sequence. What changes is whether the intermediate payoff of 150 Afghanis is embedded in an uncertain gamble or paid with certainty.

3.3 Utility under uncertainty

In Task 1, the respondent is indifferent between Option A and Option B at the switch point q:

$$0.5v(X) + 0.5v(Y) = qv(Y) + (1 - q)v(0).$$

With $v(0) = 0$ and $v(Y) = 1$, the utility of $X = 150$ under uncertainty is

$$v(X)_U = \frac{q - 0.5}{0.5}.$$

Interpretation: Under expected utility, this is a nonparametric measure of utility curvature. Relative to the risk-neutral benchmark $X/Y = 150/450 = 1/3$, values above one-third indicate risk aversion, and values below one-third indicate risk tolerance.

3.4 Utility under certainty

In Task 2, the respondent is indifferent between certainty X and the gamble $(q'; Y, 0)$:

$$v(X) = q'v(Y) + (1 - q')v(0).$$

With the same normalization, the utility of $X = 150$ under certainty is

$$v(X)_C = q'.$$

Interpretation: Expected utility predicts that the utility of a monetary outcome should be the same whether it is elicited under uncertainty or certainty. If $v(X)_C \neq v(X)_U$, the independence axiom is violated.

3.5 Certainty Premium

The paper defines

$$\text{Certainty Premium} \equiv v(X)_C - v(X)_U.$$

- Expected utility predicts Certainty Premium $= 0$.
- Cumulative prospect theory with standard inverse-S probability weighting often predicts Certainty Premium $<$ 0 in this design.
- Disappointment aversion and u–v preferences predict Certainty Premium > 0 because certainty has special value.
- Empirically, the paper finds a large positive Certainty Premium.

Interpretation: The Certainty Premium is measured in probability units of the high outcome. A premium of 0.37 means the sure 150 Afghanis is valued like roughly 37 percentage points more chance of winning 450 Afghanis relative to its uncertain utility value.

3.6 Psychological primes

- Respondents are randomly assigned to FEAR, HAPPY, or NEUTRAL recollection primes.
- The FEAR prime asks respondents to recall and describe an event that made them afraid.
- The HAPPY and NEUTRAL primes are grouped as the comparison group in the main analysis because the paper finds no meaningful differences between them.

- Randomization is stratified at the polling-center level.
- Random assignment is implemented using a fixed sequence with a random seed by polling center, reducing the risk of enumerator manipulation.

Interpretation: The prime does not create trauma. It induces recall of an emotional state. This distinction is essential because the paper studies how prior exposure interacts with current emotional recollection.

3.7 Administrative violence data

- Violence data come from ISAF CIDNE records.
- Violence is defined as at least one successful attack within one kilometer of the interview location between April 2002 and February 2010.
- Failed Violence identifies unsuccessful attacks and is used as a placebo comparison.
- The paper also tests robustness to alternative distance halos, intensity measures, vintage of violence, and self-reported personal attack experience.

Interpretation: The administrative measure reduces reliance on self-reported trauma. The failed-violence placebo asks whether the results are driven by unobserved local characteristics correlated with intended attacks rather than the realized experience of violence.

4 Models and empirical specifications

4.1 Interval regression for elicited utility

Because the switch point is observed in 10 percentage-point intervals, the dependent variable is interval-coded. The generic specification is

$$Y_i^* = \alpha + \beta \text{FEAR}_i + \gamma \text{Violence}_i + \delta(\text{FEAR}_i \times \text{Violence}_i) + X_i'\theta + \mu_p + \varepsilon_i,$$

where Y_i^* is the latent value behind one of the interval-coded outcomes:

$$Y_i \in \{v(150)_U,\ v(150)_C,\ v(150)_C - v(150)_U\}.$$

Interpretation: Interval regression respects the fact that the experiment observes a range for the utility value rather than a point estimate. Treating interval responses as exact values would overstate precision.

4.2 Main experimental-prime specification

$$Y_i^* = \alpha + \beta \text{FEAR}_i + X_i'\theta + \mu_p + \varepsilon_i.$$

- β measures the effect of fearful recollection on elicited preferences.
- The main outcomes are utility under uncertainty, utility under certainty, and the Certainty Premium.
- Province fixed effects are included in the main specifications.
- Standard errors are clustered at the polling-center level.

Interpretation: This specification identifies the average effect of fear recall, using the randomized prime.

4.3 Administrative violence specification

$$Y_i^* = \alpha + \gamma \text{Violence}_i + X_i'\theta + \mu_p + \varepsilon_i.$$

Interpretation: This is not randomized because people are not randomly assigned to violent neighborhoods. It asks whether objective local exposure alone is correlated with measured preferences.

4.4 Prime-by-violence interaction specification

$$Y_i^* = \alpha + \beta \text{FEAR}_i + \gamma \text{Violence}_i + \delta(\text{FEAR}_i \times \text{Violence}_i) + X_i'\theta + \mu_p + \varepsilon_i.$$

- δ is the key interaction coefficient.
- A positive δ in the Certainty Premium regression means that fear recall raises the value of certainty specifically among violence-exposed respondents.
- The paper finds the interaction is positive and statistically significant for the Certainty Premium.

Interpretation: The central argument is not simply "violence makes people risk averse." It is that violence-exposed individuals are especially sensitive to fear recall in certainty-related decisions.

4.5 Within-polling-center fixed effects

To address highly local confounds, the paper estimates specifications with polling-center fixed effects:

$$Y_i^* = \alpha_c + \beta \text{FEAR}_i + \delta(\text{FEAR}_i \times \text{Violence}_c) + X_i'\theta + \varepsilon_i.$$

Interpretation: Because violence is measured at the polling-center level, the main violence dummy is collinear with polling-center fixed effects. The interaction can still be identified from within-center randomized variation in FEAR.

4.6 Placebo and confound specifications

The failed-violence placebo uses

$$Y_i^* = \alpha + \beta \text{FEAR}_i + \gamma_1 \text{Violence}_i + \gamma_2 FailedViolence_i + \gamma_3(\text{Violence}_i \times FailedViolence_i) + X_i'\theta + \mu_p + \varepsilon_i.$$

The social-cohesion and migration tests interact FEAR and Violence with variables such as local dispute resolution preferences, willingness to report insurgent activity, and whether the respondent was born locally.

Interpretation: These tests ask whether the results are really about realized violence and fear recall, rather than social cohesion, selective migration, or general attack risk.

5 Identification and empirical design

5.1 What identifies the main causal effect

- The FEAR prime is randomized, so average prime effects can be interpreted causally under random assignment.

- The interaction between FEAR and Violence is identified by comparing prime effects across exposed and unexposed locations.
- Within-polling-center specifications strengthen the design by comparing individuals in the same local area who receive different primes.
- Balance tests show that demographic, social-cohesion, migration, violence-exposure, and baseline-risk variables are similar across primes.

Interpretation: The strongest identification comes from randomized fear recall. The violence dimension is observational, but the paper combines geocoded administrative data, local fixed effects, and placebo tests to make confounding less plausible.

5.2 Why the design separates exposure from triggering

- If violence permanently changed baseline preferences, Violence should predict the Certainty Premium even without FEAR.
- If fear recall only mattered generally, FEAR should affect everyone similarly.
- The paper finds the strongest effects in the FEAR $\times$ Violence interaction.
- This pattern supports a triggerability interpretation: exposure creates susceptibility to fear-induced preference shifts.

Interpretation: The empirical pattern is sharper than a simple trauma effect. It points to a state-dependent mechanism, where recall activates certainty demand among exposed respondents.

5.3 Why failed violence is a useful placebo

- Failed attacks may share many determinants of attempted violence: insurgent presence, targets, geography, political conditions, and local insecurity.
- But they do not generate the same realized violent episode as successful attacks.
- If results were driven only by local conflict risk or attempted insurgency, failed violence should predict similar effects.
- The paper finds that successful violence matters, while failed violence does not.

Interpretation: The placebo narrows the interpretation toward realized traumatic exposure rather than unobserved features of places where attacks were attempted.

5.4 Main remaining limitations

- Violence exposure is not randomized.
- The paper cannot fully observe casualties, personal trauma intensity, or exact psychological mechanisms.
- The fear prime may activate multiple emotions or memories, not only fear.
- The design measures behavior in experimental gambles, not real-world insurance or portfolio choices.
- Attrition and comprehension concerns are tested but cannot be eliminated completely.

Interpretation: The paper is strongest as evidence that fear recall causally changes certainty preference for violence-exposed individuals. It is less conclusive about the exact psychological channel or long-run welfare consequences.

6 Section-by-section study guide

6.1 Introduction

The introduction motivates a connection between trauma and economic decision making. It emphasizes that previous studies produce mixed evidence on whether disaster or conflict exposure increases or decreases risk tolerance. The paper's distinctive move is to combine objective violence exposure, experimental priming, and a preference elicitation method that can detect certainty-specific behavior.

6.2 Research design

This section is the technical core. The utility elicitation subsection derives $v(X)_U$, $v(X)_C$, and the Certainty Premium. The priming subsection explains randomized fear, happy, and neutral recall. The violence-data subsection explains how administrative attack records are mapped to polling centers.

6.3 Results

The results proceed in layers. First, the authors document positive Certainty Premia. Second, they show that FEAR and Violence alone have limited explanatory power for the main certainty result. Third, they show that FEAR $\times$ Violence robustly raises the Certainty Premium.

6.4 Robustness and alternative explanations

The robustness material addresses local confounds, failed attacks, migration, social cohesion, violence definitions, violence intensity and recency, decision errors, and attrition.

6.5 Discussion and conclusion

The conclusion interprets the results as evidence of triggerability rather than permanent baseline preference change. It also argues that policymakers and marketers may need to consider recall mechanisms when interacting with trauma-affected populations.

7 Tables and figures

7.1 Table 1: Multiple Price Lists

Read:

- Table 1 shows the two multiple price lists used to elicit risk preferences.
- In Task 1, Option A is a 50/50 gamble between 450 and 150 Afghanis; Option B is an increasing probability of 450 versus 0.
- In Task 2, Option A is a certain payment of 150 Afghanis; Option B is the same increasing probability of 450 versus 0.
- The only difference between tasks is whether the 150 Afghanis payoff is embedded in uncertainty or received for sure.

Detailed interpretation: The table is the backbone of the measurement strategy. Because the Option B lottery is identical across tasks, the switch-point difference isolates how subjects treat the same monetary amount when it is certain versus uncertain. This is why the Certainty Premium is not merely a generic risk-aversion measure.

Economic message: The table operationalizes a behavioral-economics test of the independence axiom in a field setting.

TABLE 1—MULTIPLE PRICE LISTS

$\in$	Option B	Option A
Task 1		
q'		
[0, 0.1]	10% chance of 450 Afs, 90% chance of 0 Afs	50% chance of 450 Afs, 50% chance of 150 Afs
[0.1, 0.2]	20% chance of 450 Afs, 80% chance of 0 Afs	50% chance of 450 Afs, 50% chance of 150 Afs
[0.2, 0.3]	30% chance of 450 Afs, 70% chance of 0 Afs	50% chance of 450 Afs, 50% chance of 150 Afs
[0.3, 0.4]	40% chance of 450 Afs, 60% chance of 0 Afs	50% chance of 450 Afs, 50% chance of 150 Afs
[0.4, 0.5]	50% chance of 450 Afs, 50% chance of 0 Afs	50% chance of 450 Afs, 50% chance of 150 Afs
[0.5, 0.6]	60% chance of 450 Afs, 40% chance of 0 Afs	50% chance of 450 Afs, 50% chance of 150 Afs
[0.6, 0.7]	70% chance of 450 Afs, 30% chance of 0 Afs	50% chance of 450 Afs, 50% chance of 150 Afs
[0.7, 0.8]	80% chance of 450 Afs, 20% chance of 0 Afs	50% chance of 450 Afs, 50% chance of 150 Afs
[0.8, 0.9]	90% chance of 450 Afs, 10% chance of 0 Afs	50% chance of 450 Afs, 50% chance of 150 Afs
[0.9, 1]	100% chance of 450 Afs, 0% chance of 0 Afs	50% chance of 450 Afs, 50% chance of 150 Afs
$\in$		
Task 2		
q		
[0, 0.1]	10% chance of 450 Afs, 90% chance of 0 Afs	150 Afghanis
[0.1, 0.2]	20% chance of 450 Afs, 80% chance of 0 Afs	150 Afghanis
[0.2, 0.3]	30% chance of 450 Afs, 70% chance of 0 Afs	150 Afghanis
[0.3, 0.4]	40% chance of 450 Afs, 60% chance of 0 Afs	150 Afghanis
[0.4, 0.5]	50% chance of 450 Afs, 50% chance of 0 Afs	150 Afghanis
[0.5, 0.6]	60% chance of 450 Afs, 40% chance of 0 Afs	150 Afghanis
[0.6, 0.7]	70% chance of 450 Afs, 30% chance of 0 Afs	150 Afghanis
[0.7, 0.8]	80% chance of 450 Afs, 20% chance of 0 Afs	150 Afghanis
[0.8, 0.9]	90% chance of 450 Afs, 10% chance of 0 Afs	150 Afghanis
[0.9, 1]	100% chance of 450 Afs, 0% chance of 0 Afs	150 Afghanis

TABLE 2—SUMMARY STATISTICS

	NEUTRAL prime	*FEAR* prime	*HAPPY* prime	t-test of: (p-value)	
	(1)	(2)	(3)	(2)–(1)	(3)–(1)
Sociodemographics					
Age	29.520	29.592	28.926	0.937	0.495
	(0.648)	(0.648)	(0.576)		
Income (1,000 AFs)	12.994	12.303	12.395	0.409	0.445
	(0.536)	(0.648)	(0.573)		
Female (= 1)	0.370	0.438	0.407	0.108	0.370
	(0.029)	(0.031)	(0.030)		
Shia (= 1)	0.157	0.140	0.130	0.578	0.368
	(0.022)	(0.021)	(0.020)		
Education (years)	9.719	9.796	10.004	0.834	0.442
	(0.259)	(0.261)	(0.264)		
Married (= 1)	0.626	0.619	0.593	0.858	0.418
	(0.029)	(0.030)	(0.030)		
Social cohesion and mobility					
Reporting insurgent activity important	0.496	0.525	0.474	0.521	0.637
	(0.032)	(0.032)	(0.033)		
Police resolve disputes	0.249	0.189	0.189	0.089	0.088
	(0.026)	(0.024)	(0.024)		
Courts resolve disputes	0.135	0.174	0.178	0.215	0.170
	(0.020)	(0.023)	(0.023)		
Respondent born locally	0.786	0.781	0.800	0.880	0.696
	(0.024)	(0.025)	(0.024)		
Administrative violence					
Violence (= 1)	0.473	0.464	0.485	0.689	0.656
	(0.030)	(0.030)	(0.030)		
Failed Violence (= 1)	0.363	0.355	0.344	0.841	0.650
	(0.029)	(0.029)	(0.029)		
Self-reported violence					
Self-reported attack (last five years)	0.242	0.223	0.233	0.594	0.812
	(0.026)	(0.026)	(0.026)		
Expectations of future insurgent attack (0–10)	3.491	3.438	3.544	0.818	0.820
	(0.157)	(0.172)	(0.176)		
Baseline risk					
Baseline risk (0–10)	2.246	2.015	2.296	0.263	0.810
	(0.142)	(0.149)	(0.158)		
Observations	281	265	270		

Note: Standard errors reported in parentheses.

7.2 Table 2: Summary Statistics

Read:

- Table 2 compares respondents assigned to NEUTRAL, FEAR, and HAPPY primes.
- Age, income, gender, Shia identity, education, marital status, social-cohesion measures, violence exposure, self-reported attack experience, expectations of future insurgent attack, and baseline risk tolerance are broadly balanced.
- The number of observations across prime groups is similar: 281, 265, and 270.
- The balance supports the use of randomized primes for causal identification.

Detailed interpretation: The most important role of this table is not descriptive but diagnostic. If the FEAR group differed systematically from the NEUTRAL or HAPPY groups before the prime, later differences in risk tasks could reflect selection. The table shows that such concerns are limited.

Economic message: Randomization makes emotional recall an experimental treatment. The balance table supports interpreting prime effects as caused by the recall condition rather than by preexisting group differences.

7.3 Figure 1: Successful Attacks and Unsuccessful Attacks in Kabul

Read:

- Figure 1 maps polling centers and nearby violence classifications in Kabul.
- Locations are classified as No Violence, Failed Violence, Successful Violence, or both Successful and Failed Violence.
- The figure illustrates how the authors connect geocoded administrative violence records to survey locations.
- Both successful and failed attacks are spatially dispersed, which helps motivate local fixed-effect and placebo strategies.

Detailed interpretation: The figure is not merely illustrative. It shows why geographic measurement matters. If violence were coded only at a coarse province level, exposed and unexposed respondents could be badly misclassified. The local geocoding supports sharper comparisons and permits the failed-violence placebo.

Economic message: Exposure to violence is measured as a local environmental treatment. The map shows how administrative conflict data become an empirical variable.

136 THE AMERICAN ECONOMIC REVIEW JANUARY 2014

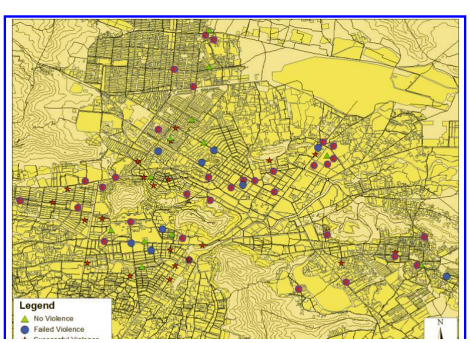

TABLE 3—SWITCH POINTS (*row number*)

		Task 2 switch point						
		4	5	6	7	8	9	10
Task 1 switch point	5	23	86	22	3			
	6		38	150	35	2	1	
	7			40	163	31	1	
	8			1	45	88	9	
	9				1	41	30	1
	10						1	2
	No switch				1			

7.4 Table 3: Switch Points (row number)

Read:

- Table 3 cross-tabulates Task 1 and Task 2 switch points for the clean sample of 816 respondents.
- A large mass of respondents switch in the same row across both tasks.
- Many respondents have Task 2 switch points that imply a higher utility value for certainty than the Task 1 utility value under uncertainty.
- The table motivates the interval-regression approach used in Tables 4–7.

Detailed interpretation: The table shows the raw empirical basis for the Certainty Premium. It also reveals why the authors avoid simply imposing a parametric risk-aversion model: the data contain direct evidence about how utility at certainty differs from utility under uncertainty.

Economic message: The raw switch-point distribution already suggests that expected utility is too restrictive for this setting.

7.5 Table 4: Attacks, Primes, and Elicited Utility

Read:

- Panel A reports the effect of the FEAR prime alone. FEAR reduces $v(150)_U$, which the authors interpret as increased risk tolerance under uncertainty.
- Panel A also shows a positive Certainty Premium of roughly 0.37 without covariates and 0.44 with covariates.
- Panel B shows that administrative Violence alone has limited direct predictive power for elicited utility.
- Panel C is the central result: FEAR×Violence raises the Certainty Premium by about 0.06, with statistical significance.
- The interaction effect is robust to adding covariates.

Detailed interpretation: This table separates three possible stories. First, fear recall could affect everyone similarly. Second, violence exposure could permanently shift preferences. Third, violence exposure could make people more sensitive to fear recall. The evidence is strongest for the third story. The direct FEAR and direct Violence effects do not by themselves explain the Certainty Premium pattern; the interaction does.

Economic message: Violence exposure changes the state-dependence of decision making. Fear recall makes certainty more valuable for exposed individuals.

7.6 Table 5: Attacks, Primes, and Elicited Utility with Polling-Center Fixed Effects

Read:

- Table 5 repeats key specifications using polling-center fixed effects.
- FEAR remains important, and the FEAR × Violence effect on the Certainty Premium remains positive.
- The interaction coefficients are somewhat smaller than in Table 4 but still economically meaningful and statistically significant.

Table 4—Attacks, Primes, and Elicited Utility

	$v(150)_u$		$v(150)_c$		$v(150)_c - v(150)_u$	
	(1)	(2)	(3)	(4)	(5)	(6)
Panel A. Priming results						
FEAR (= 1)	−0.052***	−0.068***	−0.018*	−0.024**	0.034***	0.043***
	(0.018)	(0.018)	(0.009)	(0.009)	(0.011)	(0.011)
Constant	0.256***	0.071	0.622***	0.517***	0.367***	0.442***
	(0.011)	(0.050)	(0.005)	(0.026)	(0.009)	(0.032)
Covariates	No	Yes	No	Yes	No	Yes
Observations	816	718	816	718	816	718
Clusters	278	267	278	267	278	267
log-likelihood	−1,285.164	−1,105.430	−1,300.302	−1,123.941	−572.954	−467.789
Panel B. Administrative violence results						
Violence (= 1)	−0.016	−0.011	−0.004	−0.002	0.012	0.009
	(0.015)	(0.017)	(0.007)	(0.008)	(0.011)	(0.012)
Constant	0.251***	0.057	0.620***	0.511***	0.369***	0.449***
	(0.015)	(0.052)	(0.007)	(0.027)	(0.012)	(0.033)
Covariates	No	Yes	No	Yes	No	Yes
Observations	816	718	816	718	816	718
Clusters	278	267	278	267	278	267
log-likelihood	−1,289.216	−1,112.293	−1,302.126	−1,127.166	−577.058	−474.943
Panel C. Exposure to violence and prime sensitivity						
FEAR (= 1)	−0.006	−0.023	−0.001	−0.007	0.004	0.015
	(0.025)	(0.025)	(0.012)	(0.012)	(0.016)	(0.016)
Violence (= 1)	0.015	0.019	0.007	0.009	−0.008	−0.010
	(0.019)	(0.022)	(0.009)	(0.010)	(0.014)	(0.015)
FEAR × *Violence*	−0.098***	−0.096***	−0.036**	−0.035*	0.061***	0.060***
	(0.034)	(0.035)	(0.018)	(0.018)	(0.021)	(0.022)
Constant	0.253***	0.066	0.620***	0.514***	0.367***	0.444***
	(0.017)	(0.052)	(0.008)	(0.027)	(0.014)	(0.035)
Covariates	No	Yes	No	Yes	No	Yes
Observations	816	718	816	718	816	718
Clusters	278	267	278	267	278	267
log-likelihood	−1,280.901	−1,101.704	−1,298.191	−1,122.101	−568.666	−463.932

Notes: Estimates from interval regressions (Stewart 1983). Robust standard errors clustered at the Polling Center level reported in parentheses. All regressions include province fixed effects. Violence data are from ISAF CIDNE

Table 5—Attacks, Primes, and Elicited Utility

	$v(150)_u$		$v(150)_c$		$v(150)_c - v(150)_u$	
	(1)	(2)	(3)	(4)	(5)	(6)
Panel A. Priming results						
FEAR (= 1)	−0.059***	−0.077***	−0.020**	−0.030***	0.037***	0.047***
	(0.018)	(0.018)	(0.009)	(0.009)	(0.011)	(0.010)
Constant	0.320***	0.029	0.624***	0.507***	0.305***	0.463***
	(0.084)	(0.111)	(0.043)	(0.057)	(0.051)	(0.066)
Covariates	No	Yes	No	Yes	No	Yes
Observations	816	718	816	718	816	718
log-likelihood	−1,173.515	−985.089	−1,193.460	−1,005.554	−423.597	−317.696
Panel B. Exposure to violence and prime sensitivity						
FEAR (= 1)	−0.017	−0.037	−0.007	−0.016	0.012	0.023
	(0.025)	(0.026)	(0.013)	(0.013)	(0.015)	(0.015)
FEAR × *Violence*	−0.083**	−0.079**	−0.027	−0.026	0.050**	0.047**
	(0.035)	(0.036)	(0.018)	(0.019)	(0.021)	(0.021)
Constant	0.333***	0.051	0.628***	0.514***	0.298***	0.450***
	(0.084)	(0.111)	(0.043)	(0.057)	(0.052)	(0.066)
Covariates	No	Yes	No	Yes	No	Yes
Observations	816	718	816	718	816	718
log-likelihood	−1,170.730	−982.717	−1,192.323	−1,004.462	−420.812	−315.263

Notes: Estimates from interval regressions (Stewart 1983). Standard errors reported in parentheses. All regressions include Polling Center fixed effects. There are 278 polling centers in our sample. We do not include *Violence* in panel B as it is measured at the polling center level and so is perfectly collinear with the polling center fixed effects. Violence data are from ISAF CIDNE. *Violence* is defined as a violent event occurring within one kilometer of interview location over the period April 2002–February 2010. Sample: 816 individuals with monotonic utility and no multiple switching. $v(150)_C$ refers to elicited utility under certainty, while $v(150)_U$ refers to elicited utility under uncertainty. The difference $v(150)_C - v(150)_U$ is the measured *Certainty Premium*. The covariates are preprime risk (0–10), female (= 1), Shia (= 1), years of education, born locally (= 1), reporting insurgent activity important (= 1), prefer police resolve disputes (= 1), prefer courts resolve disputes (= 1), married (= 1), age, and log(income).

- Because Violence is measured at the polling-center level, its main effect is absorbed by polling-center fixed effects.

Detailed interpretation: This table is central for addressing spatial confounding. If violent polling centers are different in unobserved ways, province fixed effects may not be enough. Polling-center fixed effects force identification to rely on the randomized prime within the same local area.

Economic message: The key result survives a tighter local comparison, strengthening the claim that the prime-exposure interaction is not merely a geographic artifact.

7.7 Table 6: Placebo Tests Using Failed Violence

Read:

- Table 6 uses failed attacks as a placebo form of local insurgent activity.
- In the FEAR subsample, successful Violence predicts a higher Certainty Premium.
- Failed Violence is not a robust predictor of the Certainty Premium.
- The interaction of Violence and Failed Violence does not overturn the successful-violence result.
- In the non-FEAR subsample, none of these violence measures strongly predicts the Certainty Premium.

Detailed interpretation: This is a clever falsification exercise. Failed attacks may capture places where insurgents attempted violence, but without the realized violent experience. If the result were just about living in high-risk neighborhoods or near insurgent targets, failed violence would have similar predictive power. It does not.

Economic message: Realized traumatic violence, not merely attempted attack risk, is what makes fear recall consequential for certainty demand.

TABLE 6—PLACEBO TESTS USING FAILED VIOLENCE

	Certainty Premium: $v(150)_C - v(150)_U$					
	FEAR (= 1)			*FEAR* (= 0)		
Sample:	(1)	(2)	(3)	(4)	(5)	(6)
Violence (= 1)	0.066***	0.054***	0.053**	−0.008	−0.007	−0.003
	(0.019)	(0.020)	(0.021)	(0.015)	(0.018)	(0.020)
Failed Violence (= 1)	0.001	−0.021	−0.023	−0.014	−0.012	0.010
	(0.019)	(0.030)	(0.030)	(0.016)	(0.030)	(0.026)
Violence × *Failed Violence*		0.036	0.035		−0.003	−0.029
		(0.038)	(0.040)		(0.032)	(0.030)
Constant	0.363***	0.367***	0.428***	0.375***	0.375***	0.450***
	(0.019)	(0.019)	(0.054)	(0.018)	(0.020)	(0.045)
Covariates	No	No	Yes	No	No	Yes
Observations	265	265	238	551	551	480
Clusters	196	196	181	258	258	239
log-likelihood	−171.215	−170.830	−141.457	−391.627	−391.621	−314.615

Notes: Estimates from interval regressions. Robust standard errors clustered at the Polling Center level reported in parentheses. All regressions include province fixed effects. Violence data are from ISAF CIDNE. *Violence* is defined as a violent event occurring within one kilometer of interview location over the period April 2002–February 2010. Sample: 816 individuals with monotonic utility and no multiple switching. The difference $v(150)_C - v(150)_U$ is the measured *Certainty Premium*. The covariates are preprime risk (0–10), female (= 1), Shia (= 1), years of education, born locally (= 1), reporting insurgent activity important (= 1), prefer police resolve disputes (= 1), prefer courts resolve disputes (= 1), married (= 1), age, and log(income).

TABLE 7—SOCIAL COHESION AND SELECTIVE MIGRATION

	$v(150)_c - v(150)_u$							
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
FEAR (= 1)	0.013	0.016	0.008	0.010	0.006	0.010	−0.023	−0.021
	(0.020)	(0.019)	(0.017)	(0.017)	(0.017)	(0.017)	(0.028)	(0.028)
Violence (= 1)	−0.010	−0.010	−0.008	−0.008	−0.008	−0.008	−0.007	−0.007
	(0.015)	(0.015)	(0.014)	(0.014)	(0.014)	(0.014)	(0.014)	(0.014)
FEAR × *Violence*	0.063***	0.060***	0.061***	0.057***	0.063***	0.058***	0.063***	0.059***
	(0.022)	(0.022)	(0.021)	(0.021)	(0.021)	(0.021)	(0.022)	(0.021)
Report. ins. act. imp.	−0.008	−0.008						
	(0.013)	(0.013)						
Police solve disputes (= 1)			0.014	0.013				
			(0.015)	(0.015)				
Court solve disputes (= 1)					0.005	0.009		
					(0.018)	(0.018)		
Born locally (= 1)							−0.026	−0.022
							(0.018)	(0.017)
FEAR × report	0.000	−0.003						
	(0.021)	(0.020)						
FEAR × police			−0.018	−0.012				
			(0.029)	(0.028)				
FEAR × court					−0.015	−0.018		
					(0.031)	(0.031)		
FEAR × local							0.034	0.034
							(0.028)	(0.028)
Constant	0.367***	0.440***	0.365***	0.421***	0.367***	0.425***	0.386***	0.439***
	(0.015)	(0.031)	(0.014)	(0.032)	(0.014)	(0.031)	(0.020)	(0.035)
Covariates	No	Yes	No	Yes	No	Yes	No	Yes
Fixed effects	Prov.	Prov.	Prov.	Prov.	Prov.	Prov.	Prov.	Prov.
Observations	718	718	816	816	816	816	816	816
Clusters	267	267	278	278	278	278	278	278
log-likelihood	−474.679	−464.557	−568.243	−558.992	−568.530	−559.183	−567.209	−558.193

Notes: Estimates from interval regressions (Stewart 1983). Robust standard errors clustered at the Polling Center level reported in parentheses. All regressions include province fixed effects. Violence data are from ISAF CIDNE. *Violence* is defined as a violent event occurring within one kilometer of interview location over the period April 2002–February 2010. Sample: 816 individuals with monotonic utility and no multiple switching. The difference

7.8 Table 7: Social Cohesion and Selective Migration

Read:

- Table 7 tests whether social cohesion or selective migration explain the FEAR × Violence result.
- The table adds measures related to reporting insurgent activity, using police or courts to resolve disputes, and being born locally.
- It also interacts these variables with FEAR.
- Across specifications, FEAR × Violence remains positive and statistically significant.
- The magnitudes remain close to those in the main specification.

Detailed interpretation: This table addresses an important selection concern. Perhaps violence-affected locations differ because migrants, socially cohesive households, or respondents trusting formal institutions sort differently across neighborhoods. The table shows that adding these variables and their FEAR interactions does not explain away the main interaction.

Economic message: The fear-triggered certainty effect is not reducible to social cohesion, migration, or dispute-resolution preferences.

8 Result map

- **Measurement result.** A two-task uncertainty-equivalent method produces separate values for $v(150)_U$ and $v(150)_C$.
- **Non-EU result.** The average Certainty Premium is positive and large, contradicting expected utility and standard cumulative prospect theory predictions in this design.
- **Prime result.** Fear recall affects elicited risk/certainty preferences, but the most important result is the interaction with violence exposure.
- **Exposure result.** Administrative Violence alone is not the whole story; it does not mechanically predict large baseline shifts in the same way.
- **Interaction result.** Fear recall among violence-exposed individuals raises the Certainty Premium.
- **Local robustness result.** The interaction survives polling-center fixed effects.
- **Placebo result.** Failed Violence does not mimic the successful violence effect.
- **Selection robustness result.** Social cohesion and migration controls do not explain the interaction.
- **Mechanism interpretation.** Trauma exposure appears to increase vulnerability to fear-triggered certainty demand rather than permanently shifting all risk choices.

9 Short summary

- The paper studies whether exposure to violence affects economic risk preferences in Afghanistan.
- It combines experimental priming, field utility elicitation, and geocoded administrative violence records.
- The experimental procedure elicits $v(150)_U$ under uncertainty and $v(150)_C$ under certainty.
- The difference $v(150)_C - v(150)_U$ is the Certainty Premium.
- The average Certainty Premium is large and positive.
- Fear recall alone changes elicited preferences, but the central effect occurs for violence-exposed respondents.
- The interaction between FEAR and Violence raises the Certainty Premium by about 0.06.
- The result survives covariates, polling-center fixed effects, failed-violence placebo tests, and social-cohesion/migration controls.
- The evidence suggests that violence exposure changes susceptibility to emotional triggers rather than simply shifting baseline risk preferences.
- The policy implication is that trauma-affected populations may respond strongly to framing, recall, or product features that highlight certainty or control.

10 Conclusion

10.1 Core conclusion

Callen, Isaqzadeh, Long, and Sprenger show that violence exposure interacts with fear recall to increase the value individuals place on certainty. The key result is not that violence-exposed individuals are always more risk averse. Rather, they become more likely to display a certainty preference when fear is made salient.

10.2 Economic intuition

Fear is associated with uncertainty and lack of control. A sure payoff removes uncertainty and restores control, so it may become especially attractive when fear is recalled. For individuals with prior violence exposure, fear recall appears more behaviorally powerful, causing a larger shift toward certainty.

10.3 Contribution to risk-preference measurement

The paper demonstrates that risk preferences are not fully captured by a single curvature parameter. Choices at certainty can behave differently from choices under uncertainty. This matters because many economic products, such as insurance and guaranteed savings products, are explicitly about certainty rather than risk in general.

10.4 Contribution to conflict and development economics

The paper gives a mechanism for why violence may affect economic behavior even after the violent event has passed. Trauma may not simply reduce wealth or alter beliefs; it may change how individuals respond to cues and frames.

10.5 Policy implications

Post-conflict policy design should consider emotional triggers. Programs involving insurance, savings, credit, compensation, or resettlement may generate different take-up depending on whether fear, safety, or certainty is made salient. Conversely, marketers or political actors could exploit certainty demand among trauma-affected populations.

10.6 Limitations

The violence measure is observational, and the exact psychological channel is not fully identified. The paper cannot directly measure fear intensity, perceived control, PTSD symptoms, or changes in beliefs caused by the prime. The outcomes are experimental choices, not observed long-run financial behavior.

10.7 Takeaway

The paper's main lesson is that violence exposure can make economic preferences state-dependent. The most important object is not baseline risk aversion, but the interaction between past trauma, current recall, and the decision environment.